Assignment 3 - Nonlinear dynamical systems

Table of Contents

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Question 1

Homogeneous steady states follow from the given parameter restrictions and the system of equations

$$0 = v(wv - \gamma),$$

$$0 = \alpha - w - wv^2.$$

The first equilibrium is

$$E_1 = (0, \alpha)$$

and always exists.

The first equation gives that $wv = \gamma$, which is bigger than zero by assumption. This implies that $w \neq 0$. Hence, $v = \frac{\gamma}{w}$. Plugging this into the second equation yields that $0 = w^2 - \alpha w + \gamma^2$. Solving this quadratic equation gives two other equilibria

$$E_2 = \left(\frac{2\gamma}{\alpha + \sqrt{\alpha^2 - 4\gamma^2}}, \frac{\alpha}{2} + \frac{1}{2}\sqrt{\alpha^2 - 4\gamma^2}\right) \quad and \quad E_3 = \left(\frac{2\gamma}{\alpha - \sqrt{\alpha^2 - 4\gamma^2}}, \frac{\alpha}{2} - \frac{1}{2}\sqrt{\alpha^2 - 4\gamma^2}\right).$$

It follows from $0 < 4\gamma^2$ that $\frac{\alpha}{2} - \frac{1}{2}\sqrt{\alpha^2 - 4\gamma^2} \geq 0$.

The equilibria E_2 and E_3 exist iff $\alpha \geq 2\gamma$ because v and w must be real. This means that only if the rainfall rate α is at least twice as big as the death rate γ, then we have all three homogeneous equilibria. Otherwise, we have only the equilibrium E_1. This supports statements $S1$ and $S2$.

Question 2

$$\mathcal{L}(v_*, w_*) = \begin{pmatrix} \partial_x^2 - \gamma + 2w_*v_* & v_*^2 \\ -2w_*v_* & \beta\partial_x - 1 - v_*^2 \end{pmatrix}$$

Question 3

Plugging in the vegetative equilibrium gives

$$\mathcal{L}(E_3) = \begin{pmatrix} \partial_x^2 + \gamma & \frac{2\gamma^2}{\alpha^2 - 2\gamma^2 - \alpha\sqrt{\alpha^2 - 4\gamma^2}} \\ -2\gamma & \beta\partial_x - 1 - \frac{2\gamma^2}{\alpha^2 - 2\gamma^2 - \alpha\sqrt{\alpha^2 - 4\gamma^2}} \end{pmatrix}.$$

Notice that $$v_* = \frac{2\gamma}{\alpha - \sqrt{\alpha^2 - 4\gamma^2}} = \frac{2\gamma\left(\alpha + \sqrt{\alpha^2 - 4\gamma^2}\right)}{\left(\alpha - \sqrt{\alpha^2 - 4\gamma^2}\right)\left(\alpha + \sqrt{\alpha^2 - 4\gamma^2}\right)} = \frac{\alpha + \sqrt{\alpha^2 - 4\gamma^2}}{2\gamma} = \mu(\alpha, \gamma)$$.

Hence, the ansatz $\varphi(x,t) = \exp(\lambda t + ikx)\psi$ gives $M\psi = \lambda\psi$ with M as in Assignment 3. The matrix M has the eigenvalues $\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\delta}}{2}$, where $\tau = \gamma - k^2 - 1 - \mu(\alpha,\gamma)^2 + i\beta k$ is the trace of M and $\delta = \gamma\mu(\alpha,\gamma)^2 - \gamma + i\beta k\gamma + k^2 + k^2\mu(\alpha,\gamma)^2 - i\beta k^3$ is the determinant of M.

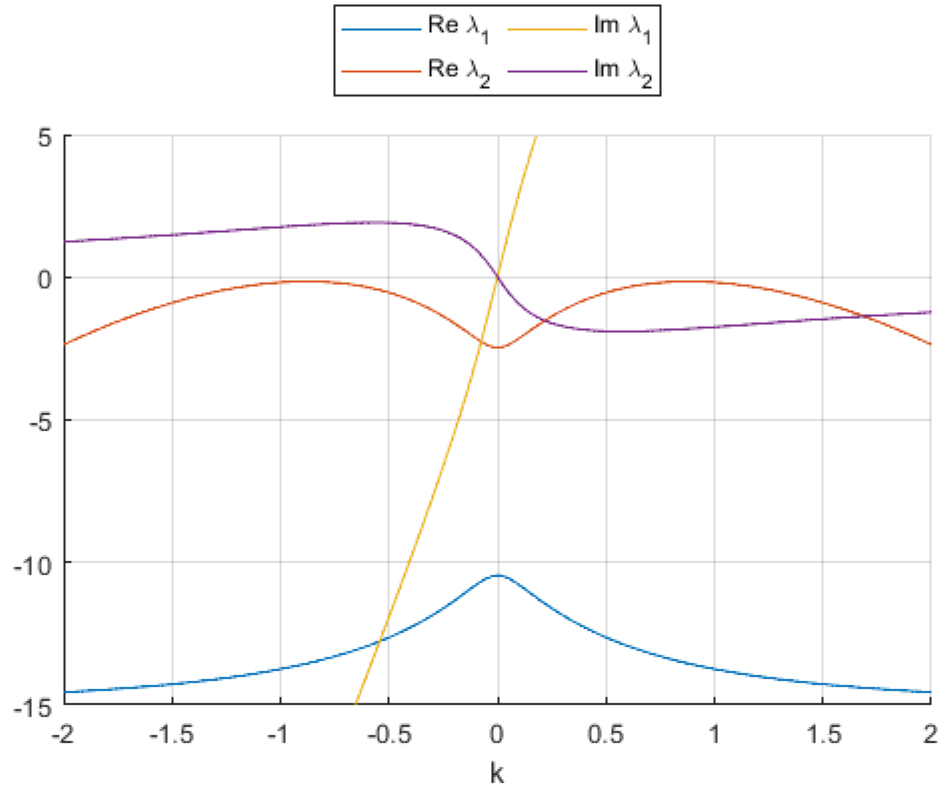
Question 4

I expected the eigenvalues to be complex conjugates. It is not the case, as it can be seen in the plot below, because the characteristic polynomial of the matrix M does not have real coefficients. This shows that a matrix with complex entries does not necessarily have complex conjugate pairs of eigenvalues.

The plot below provides a numerical evidence that the real parts of both eigenvalues for the given parameters are negative. Hence, the vegetative equilibrium is linearly stable.

```
% Trace, determinant, and eigenvalues function handles
mu = @(a,g) (a + sqrt(a^2 - 4*g^2))/(2*g);
tau  = @(a,b,g,k) g - k.^2 - 1 - mu(a,g)^2 + 1i*b*k;
delta = @(a,b,g,k) g*mu(a,g)^2 - g + 1i*b*g*k + k.^2 + mu(a,g)^2*k.^2
 - 1i*b*k.^3;
lambda1 = @(a,b,g,k) (tau(a,b,g,k)-sqrt(tau(a,b,g,k).^2 -
 4*delta(a,b,g,k)))/2;
lambda2 = @(a,b,g,k) (tau(a,b,g,k)+sqrt(tau(a,b,g,k).^2 -
 4*delta(a,b,g,k)))/2;

rel1 = @(a,b,g,k) real(lambda1(a,b,g,k));
rel2 = @(a,b,g,k) real(lambda2(a,b,g,k));
imag1 = @(a,b,g,k) imag(lambda1(a,b,g,k));
imag2 = @(a,b,g,k) imag(lambda2(a,b,g,k));

% Set parameters
a = 8; b = 20; g = 2;

% Continuous variable k
k = linspace(-2,2,1000);

% Plot dispersion relation (with inset around k=0)
figure, hold on;
plot(k,rel1(a,b,g,k),'DisplayName','Re \lambda_1');
 plot(k,rel2(a,b,g,k),'DisplayName','Re \lambda_2');
```

```
plot(k,imag1(a,b,g,k),'DisplayName','Im \lambda_1');
 plot(k,imag2(a,b,g,k),'DisplayName','Im \lambda_2');
hold off; grid on; ylim([-15 5]); xlabel('k');
lgd = legend; lgd.Location = 'northoutside'; lgd.NumColumns = 3;
drawnow;
```



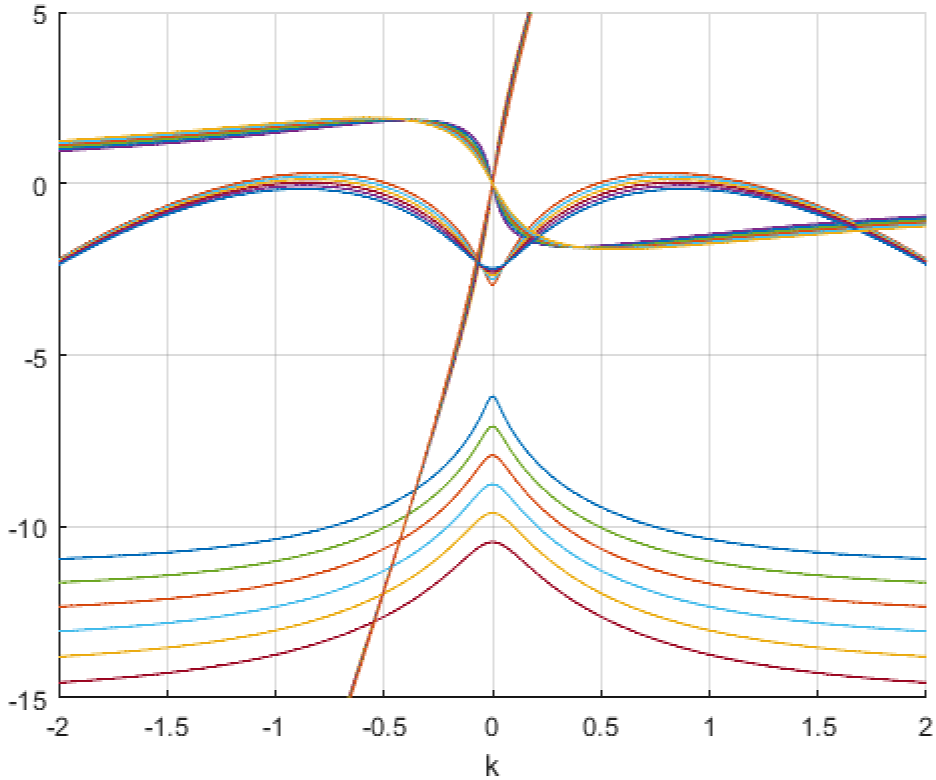
Question 5

We can see in the plot below that for $\alpha = 7$, there exist perturbations for k such that the real part of one of the corresponding eigenvalues is positive. Such system is unstable and will exhibit patterns. The plot for $\alpha = 8$ contains only eigenvalues with negative real parts. Then the system is stable and does not exhibit patterns for any perturbation. The plot provides numerical evidence that there is a critical α_c between 7 and 8 changing the stability of the vegetative equilibrium and causing bifurcation.

This plot provides numerical evidence that the bifurcation is neither Turing (there is no k s.t. both real and imaginary parts of the eigenvalues are zero) nor Hopf (at k=0, the eigenvalues are not purely imaginary). However, we can predict from Lecture notes 3 Case 4, that the pattern is a travelling wave because of the similarity of plots. This coincides with the expected pattern of a wave from the Turing bifurcation.

```
b = 20; g = 2;

figure, hold on;
for a = 7: 0.2: 8
    plot(k,rel1(a,b,g,k));              plot(k,rel2(a,b,g,k));
    plot(k,imag1(a,b,g,k));             plot(k,imag2(a,b,g,k));
end
```

```
hold off; grid on; ylim([-15 5]); xlabel('k');
drawnow;
```

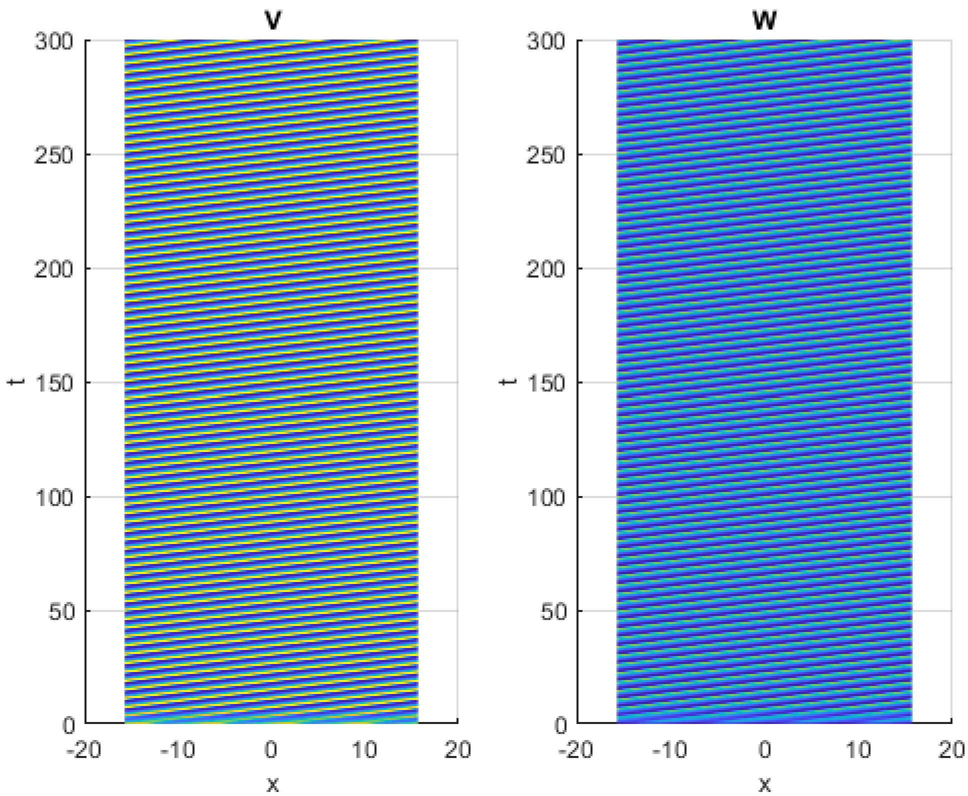


Question 6

The plot shows that the concentration of vegetation v forms a (travelling) wave pattern over time for the given parameters. This coincides with the suggested wave in Question 5. This wave pattern means that the vegetation concentration at a certain position is expected to oscillate as the time passes.

```
% Set parameters
p = [7 20 2];
L = 15.7;

% Instantiating periodic differentiation matrix
nx = 1500; [x,Dx,Dxx] = PeriodicDiffMat([-L,L],nx);

% Initial condition (steady state + perburbation)
e = ones(size(x));
v0 = 4/(7 - sqrt(33)); w0 = 7/2 - sqrt(33)/2; z0 = [v0*e; w0*e];
z0 = z0 + [cos(4*pi/L*x); e];

% Time step
rhs = @(t,z) Schnakenberg(z,p,Dx,Dxx);
jac = @(t,z) SchnakenbergJacobian(z,p,Dx,Dxx);
opts = odeset('Jacobian',jac);
tSpan = 0:0.1:300;
```

```
% tSpan = [0:0.1:50];
[t,ZHist] = ode15s(rhs,tSpan,z0,opts);

% Space-time plot
PlotHistory(x,t,ZHist,p,[]);
```



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